$\frac{}{210N}$ $\frac{}{70}$ $= 1.43\%$

30 In medical radiotherapy, different types of radioactive emissions are chosen based on their penetrating abilities and ionising effects. Which combination correctly matches the emission type with its most appropriate medical application?

A alpha particles - external beam therapy for deep tumours

B beta particles - treating shallow skin cancers and surface tumours

C gamma rays - treating only surface-level conditions

D alpha particles - medical imaging of internal organs

Ans: B

A: This is incorrect because alpha particles have extremely poor penetrating ability and are completely stopped by the skin's outer layer. They cannot reach deep tumours from an external source, making external beam therapy impossible. All the radiation would be absorbed at the skin surface without reaching the intended target.
B: This is correct because beta particles possess moderate penetrating ability, allowing them to penetrate several millimetres into tissue whilst delivering effective ionising radiation to surface and near-surface tumours. Their limited range prevents excessive damage to deeper healthy tissues, making them ideal for treating skin cancers and shallow tumours.
C: This is incorrect because gamma rays have very high penetrating ability and would largely pass through surface tissues without depositing sufficient energy for treatment. Using such penetrating radiation for surface conditions would expose deeper healthy tissues to unnecessary radiation whilst providing poor treatment efficacy at the target site.
D: This is incorrect because alpha particles cannot penetrate body tissues from external sources and cannot exit the body to reach detectors when administered internally. Medical imaging requires radiation that can either penetrate the body or travel from internal sources to external detectors, which alpha particles cannot accomplish due to their extremely limited range.